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Optimizing Economic Dispatch for Microgrid Clusters Using Improved Grey Wolf Optimization

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Abstract: With the rapid development of renewable energy generation in recent years, microgrid technology has increasingly emerged as an effective means to facilitate the integration of renewable energy. To efficiently achieve optimal scheduling for microgrid cluster (MGC) systems while guaranteeing the safe and stable operation of a power grid, this study, drawing on actual electricity-consumption patterns and renewable energy generation in low-latitude coastal areas, proposes an integrated multi-objective coordinated optimization strategy. The objective function includes not only operational costs, environmental costs, and energy storage losses but also introduces penalty terms to comprehensively reflect the operation of the MGC system. To further enhance the efficiency of solving the economic dispatch model, this study combines chaotic mapping and dynamic opposition-based learning with the traditional Grey Wolf Optimization (GWO) algorithm, using the improved GWO (CDGWO) algorithm for optimization. Comparative experiments comprehensively validate the significant advantages of the proposed optimization algorithm in terms of economic benefits and scheduling efficiency. The results indicate that the proposed scheduling strategy, objective model, and solution algorithm can efficiently and effectively achieve multi-objective coordinated optimization scheduling for MGC systems, significantly enhancing the overall economic benefits of the MGC while ensuring a reliable power supply.



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Keywords: microgrid cluster (MGC); grey wolf optimization (GWO); optimal scheduling; multi-objective optimization; economic dispatch

1. Introduction

To address the current fossil energy crisis and increasingly severe climate issues, there is a global consensus on vigorously developing renewable energy sources. Over the past few years, the application of microgrid technology has become increasingly widespread, aiming to promote the integration of renewable energy, enhance the flexibility and stability of power system structures, and ensure power supply reliability [1–5].

Distributed renewable energy generation often exhibits characteristics of decentralized concentration, a proximity to users, and small-scale deployment [6–8]. This is especially true for urban coastal areas with lower latitudes, where wind and solar resources are abundant, allowing for the dense placement of small-scale microgrids. Coordinated scheduling among multiple adjacent microgrids, often referred to as a microgrid cluster (MGC), can facilitate the local integration of renewable energy sources. This coordination within a microgrid cluster is crucial for environmental protection and alleviating pressure on remote grid supplies [9–12].

Given the rapid development of renewable energy and ongoing advancements in smart grid technologies, the coordinated operation and scheduling of MGCs have emerged as pivotal

research avenues [1,12]. This issue spans multiple layers and dimensions, thereby presenting considerable complexity. While various methods have been proposed and applied to address practical challenges in this domain, significant hurdles remain to be overcome.

Identifying the key constraints in the power scheduling problem of the MGC system is essential for maintaining the reliability and efficiency of the entire power system. Common constraints include power balance constraints and equipment constraints. Power balance constraints in the MGC system ensure that the total power generation matches the total power demand at all times. By adhering to power balance constraints, the system can avoid the extremes of power supply shortages and an oversupply of power [13–15]. Equipment constraints ensure the safe and stable operation of the system while extending the lifespan of various components. Common equipment constraints include the output power and ramping constraints for conventional thermal power generation [13,16], as well as the charging/discharging power, capacity, and State-of-Charge (SOC) constraints for energy storage systems (ESSs) [16–18]. Understanding and managing these constraints is crucial for optimizing MGC performance.

Constructing a reasonable objective function is central to the economic dispatch problem. The recent literature on MGC systems often constructs objective functions focusing on operational and environmental costs [19–21]. The calculation of operational costs in MGC systems resembles that in conventional microgrids and typically includes net costs after transactions between each MG and the main grid, net costs after transactions between MGs, maintenance costs for WT and PV generation, and generation costs for non-renewable energy sources. Environmental costs focus on treatment expenses for pollutants such as CO₂, SO₂, and NO_x from thermal power generation. This paper attempts to enhance the commonly used objective functions by introducing more factors, providing a comprehensive representation of the MGC system's economic costs and power quality.

Finding the optimal solution method for the economic dispatch model of MGCs is a key focus of many research papers in the related field, and numerous optimization algorithms have been applied to this area. Although classical Linear Programming (LP) [22] and Nonlinear Programming (NLP) [23] are simple and convenient, they are no longer suitable for the increasingly complex structure of modern power systems [24–26]. In recent years, Machine Learning has become increasingly prevalent in addressing scheduling challenges within microgrids (MGs) and MGCs [27]. Deep learning algorithms like Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Deep Reinforcement Learning (DRL) autonomously acquire operational strategies for MGs to adapt to environmental shifts and market dynamics [27–29]. However, they demand substantial volumes of training data, extensive training periods, and computational resources [30].

Alongside classical and Machine Learning algorithms, metaheuristic algorithms have found expanded utility in MGC systems. For instance, the Genetic Algorithm (GA) proves adept at tackling multi-objective optimization problems, commonly utilized for navigating the trade-offs between minimizing costs and emissions in distribution network operations. Although offering high flexibility, this algorithm demands precise parameter initialization and risks getting trapped in local optima. Particle Swarm Optimization (PSO), as a prevalent classical intelligent algorithm, stands out for its straightforward implementation and rapid convergence. However, its pursuit of the optimal solution tends to be singular, resulting in unstable performance in high-dimensional and complex problem spaces [31–34]. The Firefly Algorithm (FA) is known for its robust global search capability, making it suitable for various optimization problems. However, it can struggle with high-dimensional problems and has longer iteration times. Similarly, the Whale Optimization Algorithm (WOA) excels in global search and fast convergence with fewer parameters to adjust, making it user-friendly. Nevertheless, it still has the risk of getting trapped in local optima, and its performance can be highly influenced by the initial population quality and specific parameters.

The Grey Wolf Optimization (GWO) algorithm is renowned for its simplicity, ease of implementation, and strong global search capabilities. By effectively balancing exploration

and exploitation phases, GWO is particularly suitable for complex optimization problems. These qualities make GWO an excellent candidate for optimizing the economic dispatch in MGCs. Consequently, this study enhances the GWO algorithm and applies it to solve optimization problems in this context.

This study constructs the structure of an MGC system and comprehensively investigates the constraints on the equipment, energy storage, and power balance within each MG. To achieve an optimal economic dispatch while maintaining power quality and extending the lifespan of the MGC system's equipment, penalty terms are introduced into the objective function alongside various costs. This approach aims to balance the economic efficiency and stability of the power system. Finally, the traditional GWO algorithm is enhanced to improve its convergence speed and global search capabilities, making it more suitable for optimizing the economic dispatch of MGCs.

The major contributions of this study can be summarized as follows:

1. This research develops the innovative integration of chaos mapping and a dynamic opposition-based learning strategy with GWO (CDGWO), significantly improving the performance of the optimization algorithm. Comparative experiments confirm the notable advantages of the improved algorithm in solving the economic dispatch problem of MGCs.
2. This study includes penalty terms in the objective function to account for power exchange limits between the main grid and the MGC and for discrepancies in energy levels between the start and end of the ESS operating cycle. This integration enhances the reliability of the power supply in the MGC system and the longevity of the microgrid components.

2. Structure and Economic Dispatch Model of the Microgrid Cluster System

2.1. Microgrid Cluster System Structure

The MGC analyzed in this study consists of three independent microgrids (MGs), each equipped with key components such as wind turbines (WTs), Photovoltaic (PV) arrays, microturbines (MTs) or Diesel Generators (DGs), battery energy storage systems (ESSs), and Alternating Current (AC) loads for end-users [35,36].

The Energy Management Center (EMC) plays a crucial role in orchestrating the energy exchange among these MGs, as well as handling the transactions for buying and selling electricity with the main grid. This centralized coordination ensures optimal energy distribution and enhances the overall efficiency and reliability of the microgrid network [3,37–40].

Figure 1 illustrates the structure of the MGC system. The specific configuration of its sub-microgrid MG2 is detailed in this figure. The structures of MG1 and MG3 are essentially identical to MG2, except that the non-renewable energy generation equipment in these two sub-microgrids is not microturbines (MTs) but diesel generators (DGs). This variation in non-renewable energy generation equipment allows for a comparison of performance and operational strategies across different energy source types within the MGC.

2.2. Constraints of Microgrid Cluster System

2.2.1. Power Balance Constraints

Mathematically, the power balance constraint can be expressed as (1).

$$\sum_{i=1}^N P_{gen,i}(t) + P_{import}(t) = P_{load}(t) \quad (1)$$

$P_{gen,i}(t)$ represents the power output of the i -th generation unit at time t . $P_{import}(t)$ denotes the power imported from the main grid at time t .

The MG i is as shown in (2).

$$P_{loadi}(t) = P_{WTi}(t) + P_{PVi}(t) + P_{MTi}(t) + P_{ESSi}(t) + P_{buyi}(t) - P_{selli}(t) + P_{exi-j}(t) + P_{exi-j}(t) \quad (2)$$

The meanings of the symbols used in (2) are detailed in Table 1.

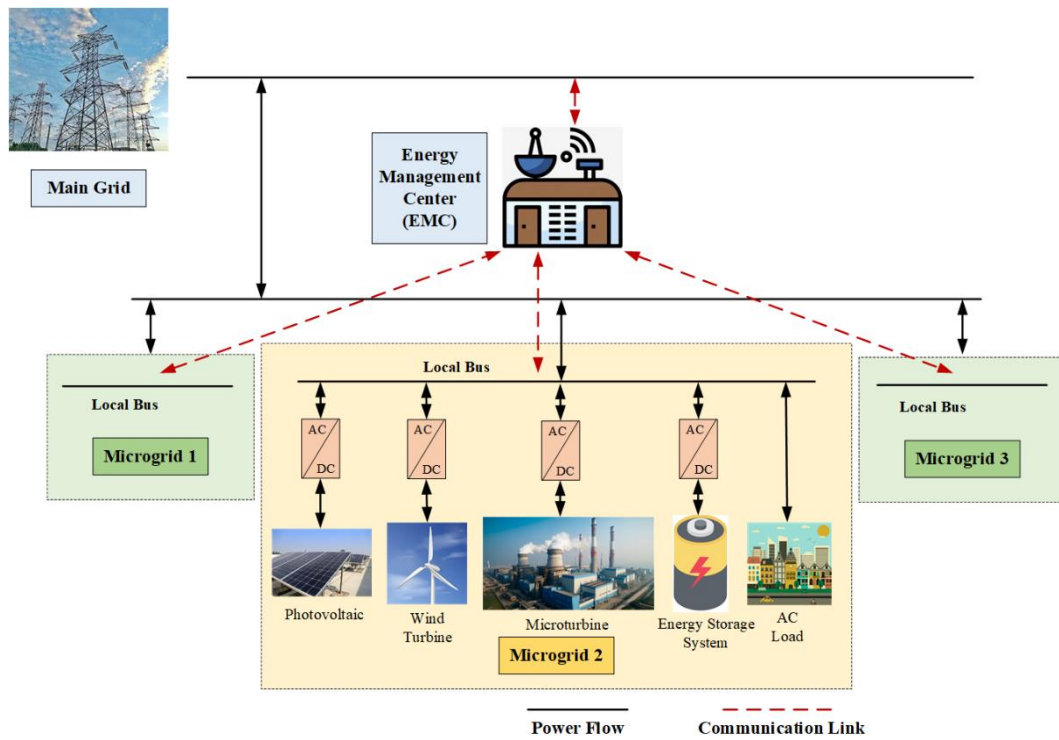


Figure 1. The structure of the microgrid cluster (MGC) system.

Table 1. Definition of symbols in the power balance constraint equation for each MG.

Symbol	Definition
$P_{loadi}(t)$	Total load demand of the MG- i at time t .
$P_{WTi}(t)$	Wind turbine power output of the MG- i at time t .
$P_{PVi}(t)$	Photovoltaic power output of the MG- i at time t .
$P_{MTi}(t)$	Power output of the microturbine in the MG- i at time t .
$P_{DGi}(t)$	Power output of the diesel generator in the MG- i at time t .
$P_{ESSi}(t)$	Power output of the ESS in the MG- i at time t .
$P_{buyi}(t)$	Power purchased from the main grid by MG- i at time t .
$P_{sell i}(t)$	Power sold to the main grid by MG- i at time t .
$P_{exi-j}(t)$	The power exchange between MG- i and MG- j at time t .

2.2.2. Equipment Self-Constrains

- Constraints on MT and DG

The Constraints on the output power for MTs and DGs are illustrated in (3).

$$\begin{cases} P_{MT}^{\min}(t) \leq P_{MT}(t) \leq P_{MT}^{\max}(t) \\ P_{DG}^{\min}(t) \leq P_{DG}(t) \leq P_{DG}^{\max}(t) \end{cases} \quad (3)$$

The ramping constraints for MTs and DGs are illustrated in (4).

$$\begin{cases} P_{MT}(t) - P_{MT}(t-1) \leq r_{MT} \\ P_{DG}(t) - P_{DG}(t-1) \leq r_{DG} \end{cases} \quad (4)$$

- Constraints on ESS

The Charging/Discharging Power Constraints of the ESS are as illustrated in (5).

$$\begin{cases} 0 \leq P_{ch}(t) \leq P_{ch}^{\max}(t) \\ 0 \leq P_{dis}(t) \leq P_{dis}^{\max}(t) \end{cases} \quad (5)$$

where $P_{ch}(t)$ and $P_{dis}(t)$ represent the charging and discharging power at time t , $P_{ch}^{max}(t)$ and $P_{dis}^{max}(t)$ is the maximum charging and discharging power limit.

The Capacity Constraint of the ESS are as illustrated in (6).

$$E_{min} \leq E(t) \leq E_{max} \quad (6)$$

where E_{min} and E_{max} set the upper and lower limits of the ESS capacity.

Effective SOC management strategies are employed to maintain the ESS within optimal operating ranges and to extend the battery life, as illustrated in (7).

$$SOC^{min} \leq SOC(t) \leq SOC^{max} \quad (7)$$

The minimum state of charge (SOC^{min}) is often set at around 20–30% to ensure there is always some reserve capacity in the battery, which helps in emergencies and prevents deep discharging, while the maximum state of charge (SOC^{max}) is typically set at 90–95% to avoid overcharging, which can lead to overheating and battery degradation. The SOC of the ESS in this study is required to be maintained between 30% and 90%.

2.3. Construction of the Objective Function

Most studies on microgrid economic planning focus on constructing objective functions based on operational and environmental costs. To improve the accuracy and comprehensiveness of the analysis, this paper adds ESS Loss Costs to these considerations. Besides these three costs, this study also incorporates two penalty terms into the objective function to enhance the power quality of the MGC and extend the lifespan of the ESS. These penalties are essential for maintaining system stability and efficiency.

Based on the above discussion, the objective function is constructed as shown in (8).

$$F_{MGC} = C_{Operation} + C_{Pollution} + C_{ESS} + F_{Main-MGC} + F_{ESS} \quad (8)$$

where $C_{Operation}$, $C_{Pollution}$, and C_{ESS} denote the functions corresponding to Operational Costs, Pollution Control Costs, and ESS Loss Costs. $F_{Main-MGC}$ and F_{ESS} are penalty functions, representing the penalty for power-exchange-exceeding limits between the main grid and the MGC and the penalty for discrepancies in energy levels between the start and end of the ESS operating cycle.

2.3.1. Operational Costs

The operational costs of the MGC system mainly include the net expenditure for purchasing and selling electricity between each sub-microgrid and the main grid, the net expenditure for purchasing and selling electricity between sub-microgrids, the maintenance costs for WT and PV generation, and the generation costs for non-renewable energy sources.

The specific mathematical expression is shown in (9) and (10).

$$F_{Operation} = C_{grid1} + C_{grid2} + C_{grid3} + C_{ex1-2} + C_{ex2-3} + C_{ex3-1} + \sum_{i=1}^3 C_{WTi} + \sum_{i=1}^3 C_{PVi} + \sum_{i=1}^3 C_{non-renew-i} \quad (9)$$

$$C_{gridi} = C_{buyi} - C_{sell_i} \quad i = 1, 2, 3 \quad (10)$$

2.3.2. Pollution Control Costs

The MGC system explored in this study has not achieved complete renewable energy integration yet. Wind turbines and photovoltaics, crucial components of the energy generation process, are notable for their absence of emissions harmful to the environment. Additionally, the environmental impact of ESSs remains minimal. This study consciously overlooks any potential pollutants that the ESS may generate, focusing instead on micro gas turbines and diesel generators, as depicted in (11).

$$F_{Pollution} = \int_0^{24} \sum_{i=1}^M (c_{DG} \cdot \lambda_i \cdot P_{DG}(t) + c_{MT} \cdot \lambda_i \cdot P_{MT}(t)) dt \quad (11)$$

where c_{DG} and c_{MT} represent the unit costs of pollution control for MTs and DGs, respectively, while λ_i denotes the emission coefficient of pollutants.

2.3.3. Energy Storage System Loss Costs

Accounting for ESS losses in cost calculations is essential for several reasons. The depth and frequency of charge and discharge cycles directly affect the lifespan of the ESS. In applications such as peak shaving and load leveling, frequent cycling can result in substantial degradation. Therefore, considering the losses of supercapacitors is crucial for the economic operation of MGCs. This consideration ensures a comprehensive evaluation of the system's long-term performance and cost-effectiveness, leading to more informed decision making and optimized resource utilization [41–43]. The corresponding mathematical model for ESS loss costs can be simplified to (12).

$$\begin{cases} F_{ESS} = m_{ESS} \sum_{i=1}^3 \int_0^{24} |P_{SCi}(t)| \cdot f(SOC_{SCi}(t)) dt \\ m_{ESS} = \frac{C_{Investment}}{Q_{ESS}} \end{cases} \quad (12)$$

where $P_{SC1}(t)$, $P_{SC2}(t)$, and $P_{SC3}(t)$ represent the power of the first and second supercapacitors, respectively. $SOC_{SC1}(t)$, $SOC_{SC2}(t)$, and $SOC_{SC3}(t)$ denote the state of charge of the first and second supercapacitors, respectively. $f(SOC_{SCi}(t))$ represents the loss function associated with the state of charge of the supercapacitor.

m_{ESS} is the unit loss cost coefficient of the energy storage system. Mathematically, it is defined as the ratio of the investment cost of the energy storage system to the total charge and discharge energy over the entire lifecycle of the supercapacitor.

2.3.4. Penalty Function for Power-Exchange-Exceeding Limits

Introducing the penalty function for a power exchange between the main grid and the MGC exceeding predefined limits serves a critical role in optimizing the objective function. It ensures an equilibrium of the power exchange between these entities, which is vital for sustaining the stability and efficiency of the overall system operation. By penalizing deviations from the desired balance, the optimization process is steered toward solutions that prioritize a smooth power transmission between the main grid and the microgrid cluster. This approach enhances the system reliability and performance by promoting a balanced and stable power exchange mechanism. The corresponding penalty function expressions for the MG and MGC are given in (13) and (14), respectively.

$$F_{Main-MGi} = \delta \sum_{t=1}^{24} (P_{MG,i}(t) - P_{ML,i}(t)) \quad i = 1, 2, 3 \quad (13)$$

$$F_{Main-MGC} = \sum_{i=1}^3 F_{Main-MGi} \quad (14)$$

where δ represents the penalty coefficient, while $P_{MG,i}(t)$ and $P_{ML,i}(t)$ denote the equivalent generation power and equivalent load power transmitted between the MGi and the distribution network at time t .

2.3.5. Penalty Function for Discrepancies in Energy Levels between Start and End of ESS Operating Cycle

The penalty term, F_{ESS} , serves as a crucial mechanism to mitigate irregularities in the charging and discharging patterns of the ESS. Penalizing deviations in ESS energy levels encourages the optimization algorithm to prioritize balanced energy usage throughout the operational cycle, enhancing the storage and distribution efficiency. This optimization

boosts the microgrid cluster system's overall performance while also extending the ESS's operational lifespan, ensuring sustained reliability within the microgrid ecosystem.

$$F_{ESS} = \gamma \left| \sum_{t, P_{dis}(t) > 0}^{24} P_{dis}(t) \eta_{dis} + \sum_{t, P_{ch}(t) > 0}^{24} P_{ch}(t) / \eta_{ch} \right| \quad (15)$$

In (15), γ represents the constraint penalty factor for the battery, while η_{dis} and η_{ch} , respectively, denote the discharging and charging efficiencies of the ESS.

3. Model Solving Method

This study improves the traditional GWO algorithm to enhance its efficiency and accuracy and applies it to solve the economic dispatch model for the MGC system described.

3.1. Traditional GWO

For the economic dispatch problem of microgrid systems, a comprehensive analysis of existing metaheuristic algorithms shows that the GWO algorithm performs well in terms of convergence speed and computational cost, demonstrating strong global search capabilities [44–46]. Building upon these advantages, this paper chooses to refine and enhance the GWO algorithm to facilitate the achievement of economic dispatch objectives in the MGC system.

Proposed by Mirjalili et al., the GWO algorithm emulates the leadership hierarchy and hunting mechanism of grey wolves in nature [44].

The GWO algorithm is founded on a fundamental mathematical framework that mirrors the social hierarchy observed in grey wolf populations. By emulating the hunting behaviors and population dynamics of grey wolves, it aims to achieve optimization objectives. The hierarchical structure is illustrated in Figure 2, with the roles and responsibilities of wolves at each level outlined in Table 2.

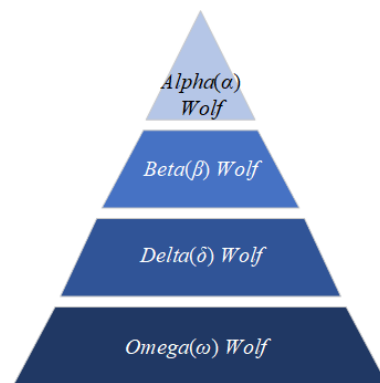


Figure 2. Hierarchy structure of wolf pack in GWO algorithm.

Table 2. The hierarchy and corresponding roles of wolf pack in GWO algorithm.

Level	Rank	Roles and Responsibilities
The first level	Alpha (α) wolf	Leader—dominates the pack.
The second level	Beta (β) wolf	Assists alpha and helps manage the pack.
The third level	Delta (δ) wolf	Follows alpha and beta and ensures tasks are carried out.
The fourth level	Omega (ω) wolf	Lowest rank—follows alpha, beta, and delta.

In the hunting sequence of grey wolves, three distinct phases are discernible: searching for prey, encircling prey, and attacking prey.

Within the framework of the GWO algorithm, the act of grey wolves surrounding prey can be conceptualized through the following position update (16).

$$\begin{cases} \vec{D} = \left| \vec{C} * \vec{X}_{prey}(t) - \vec{X}(t) \right| \\ \vec{X}(t+1) = \vec{X}_{prey}(t) - \vec{A}\vec{D} \end{cases} \quad (16)$$

where $\vec{D}$ represents the distance between individual grey wolves and the prey, $\vec{X}$ denotes the position of each individual in the population, $\vec{X}_{prey}$ stands for the position of the prey, and t signifies the iteration count. Additionally, $\vec{A}$ and $\vec{C}$ represent the vector coefficients determined by (17).

$$\begin{cases} \vec{A} = 2\vec{a} \cdot \vec{r}_1 - \vec{a} \\ \vec{C} = 2 \cdot \vec{r}_2 \end{cases} \quad (17)$$

where $\vec{r}_1$ and $\vec{r}_2$ are random vector values within the range of $[0, 1]$, and $\vec{a}$ represents the convergence factor, which governs the equilibrium between the “surrounding prey” and “attacking prey” actions in the GWO algorithm. Typically, for this parameter, it is commonly handled by linearly decreasing its value from 2 to 0 as the iteration count increases.

Following the natural hierarchy, the pursuit of prey in GWO is guided by the α wolf, with other rank wolves cooperating to encircle, pursue, and attack the prey. Accordingly, during the algorithm’s iteration, the α wolf is considered the optimal individual, followed by β and then δ individuals. Following the outlined logic and rules, the position update (18) and (19) for the grey wolf individual ω , distinct from α , β , and δ , is established.

$$\begin{cases} \vec{X}_1(t) = \left| \vec{X}_\alpha(t) - \vec{A}_1 \cdot \left| \vec{C}_1 \cdot \vec{X}_\alpha(t) - \vec{X}(t) \right| \right| \\ \vec{X}_2(t) = \left| \vec{X}_\beta(t) - \vec{A}_2 \cdot \left| \vec{C}_2 \cdot \vec{X}_\beta(t) - \vec{X}(t) \right| \right| \\ \vec{X}_3(t) = \left| \vec{X}_\delta(t) - \vec{A}_3 \cdot \left| \vec{C}_3 \cdot \vec{X}_\delta(t) - \vec{X}(t) \right| \right| \end{cases} \quad (18)$$

$$\vec{X}_\omega(t+1) = \frac{1}{3} \cdot \sum_{i=1}^3 \vec{X}_i(t) \quad (19)$$

where $\vec{X}_\alpha(t)$, $\vec{X}_\beta(t)$, and $\vec{X}_\delta(t)$ represent the positions of the α , β , and δ wolves, respectively, at time t . Additionally, $\vec{A}_1$, $\vec{A}_2$, and $\vec{A}_3$ are similar to $\vec{A}$, $\vec{C}_1$, and $\vec{C}_2$, and $\vec{C}_3$ is similar to $\vec{C}$, which is described as (17).

3.2. The Improved GWO Algorithm (CDGWO)

Traditional GWO is a nature-inspired algorithm celebrated for its simplicity and effectiveness in solving optimization problems. A key advantage of GWO is its ability to balance exploration and exploitation, making it robust across various types of optimization tasks. However, traditional GWO has its limitations, including a propensity to get trapped in local optima and slow convergence speeds, especially in complex, high-dimensional problem spaces.

To address these shortcomings, this study introduces chaos optimization and a dynamic opposition-based learning strategy. Chaos optimization helps enhance the global search ability and prevent premature convergence by introducing randomness and nonlinearity into the search process. The dynamic opposition-based learning strategy improves the convergence speed and solution accuracy by considering opposite solutions and dynamically updating them throughout the optimization process. These enhancements aim

to improve the overall performance of GWO, thereby enabling more efficient solutions for the economic dispatch challenges in the MGC system.

3.2.1. Chaos Optimization

Chaos mappings can generate sequences characterized by high randomness and nonlinearity, suitable for replacing conventional uniformly distributed random numbers in intelligent optimization algorithms [47–49]. These mappings enhance the global search efficiency and robustness of algorithms [47], reduce dependence on the selection of initial solutions and parameter settings, and offer more flexible optimization strategies for MGC scheduling problems. This paper discusses several common chaos mappings and selects the most appropriate one to enhance the GWO algorithm, tailored to the context described.

Although Tent mapping exhibits piecewise linear properties, it may lack sufficient flexibility in large-scale optimization scenarios. Sine mapping, when applied to high-dimensional optimization problems, is prone to converging to local optima, with its performance being highly dependent on the choice of parameters. In contrast, Logistic mapping is straightforward to implement and possesses robust chaotic characteristics. Chebyshev mapping maintains better uniformity and distribution within a certain range. Both mappings can significantly improve the search capabilities of the GWO algorithm, expand the diversity of the search space, and elevate the efficiency of optimization. The expressions for these mappings are presented in Table 3.

Table 3. Expressions of several commonly used chaotic maps.

Example	Expression
Tent	$x_{i+1} = \begin{cases} x_i/a & x_i < a \\ (1 - x_i)/(1 - a) & x_i \geq a \end{cases}$ $a \in (0, 1)$
Sine	$x_{i+1} = \frac{a}{4} \sin(\pi x_i)$ $a = 4$
Chebyshev	$x_{i+1} = \cos(a \cdot \cos^{-1}(x_i))$ $a = 4$
Logistic	$x_{i+1} = ax_i(1 - x_i)$ $a \in (0, 4]$

By qualitatively analyzing the characteristics of several common mappings, it can be inferred that Logistic mapping and Chebyshev mapping are better suited for the problem context outlined in this paper. To precisely select the chaotic mapping that enhances the performance of GWO, this study will empirically compare and identify the most effective chaotic mapping in Section 4.

3.2.2. Dynamic Opposition-Based Learning Strategy

The GWO algorithm employs a hierarchical strategy, with α , β , and δ wolves leading the search process. This hierarchy helps direct subordinate agents to explore the most promising areas for optimal solutions, effectively reducing the risk of premature convergence to local optima. However, the strong exploitation capability inherent in this method can limit the search diversity and can hinder the exploratory potential [48].

To overcome this limitation, this paper introduces the dynamic opposition-based learning (DOBL) strategy to enhance the GWO algorithm. Traditional opposition-based learning (OBL) [50,51] explores both current and opposite positions to improve the solution quality, but DOBL advances this concept by incorporating a dynamic factor r that changes nonlinearly with each iteration. This innovation allows for the generation of reverse

solutions to be more responsive to the search space's evolving landscape, thereby boosting the algorithm's capacity to explore diverse regions and escape local optima.

$$\begin{cases} r = \sin(t/T) \\ \tilde{X}_i(t) = pop_{max} + pop_{min} - rX_i(t), \quad i = 1, 2, \dots, n \end{cases} \quad (20)$$

Equation (20) illustrates the modified position update formula, which accounts for the dynamic factor r . Here, pop_{max} and pop_{min} define the upper and lower boundaries of the search space, respectively. $X_i(t)$ indicates the position of the i -th individual grey wolf at the t -th iteration. $\tilde{X}_i(t)$ represents the corresponding dynamic reverse solution. t denotes the iteration time, and T represents the total number of iterations.

3.2.3. The Specific Steps of the Improved GWO (CDGWO)

This paper presents an enhanced GWO algorithm for the economic dispatch of the MGC system, incorporating chaotic mapping and the dynamic opposition-based learning strategy. The detailed algorithmic procedure is illustrated in Figure 3. The specific steps of execution are outlined below.

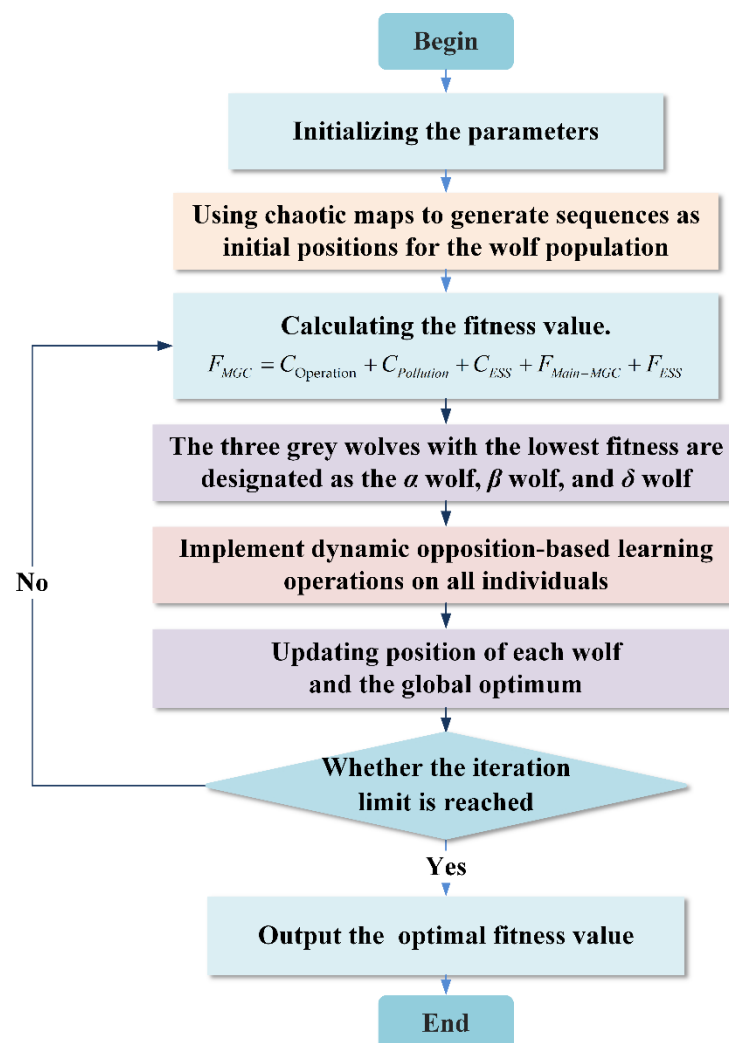


Figure 3. The procedure for the improved GWO algorithm.

Step 1: initialize the parameters of the grey wolf population.

Step 2: Use chaotic maps to generate sequences as initial positions for the wolf population. The chaotic map formulas are presented in Table 3, which includes expressions of several commonly used chaotic maps.

Step 3: Establish the fitness function, which is identical to the objective function constructed for the microgrid cluster optimization scheduling strategy, as shown in (8). Evaluate the fitness value of the entire population.

Step 4: Identify the three grey wolves with the lowest fitness values as the α wolf, β wolf, and δ wolf. These wolves will guide the rest of the population.

Step 5: perform dynamic opposition-based learning by executing searches at both the current positions and their direct opposites to increase the likelihood of finding superior solutions.

Step 6: Update individual positions based on the guidance of the α wolf, β wolf, and δ wolf. Then, update the global optimum.

Step 7: Assess whether the termination criteria have been met. If fulfilled, output the optimal fitness value. If not, recalculate the fitness value based on the updated individual positions and iterate again.

3.3. The Solution Process for the Constructed MGC Economic Dispatch Model

This paper begins by constructing the topology of the microgrid cluster and defining the system constraints necessary for safe operation. An optimal dispatch objective function is then developed to balance economic efficiency and power quality, resulting in a comprehensive multi-objective economic dispatch model. Subsequently, this study enhances the traditional GWO algorithm, resulting in the CDGWO algorithm, which features improved global search capabilities and higher optimization efficiency and accuracy. This enhanced algorithm is applied to solve the model. The specific methodological approach is illustrated in the block diagram shown in Figure 4.

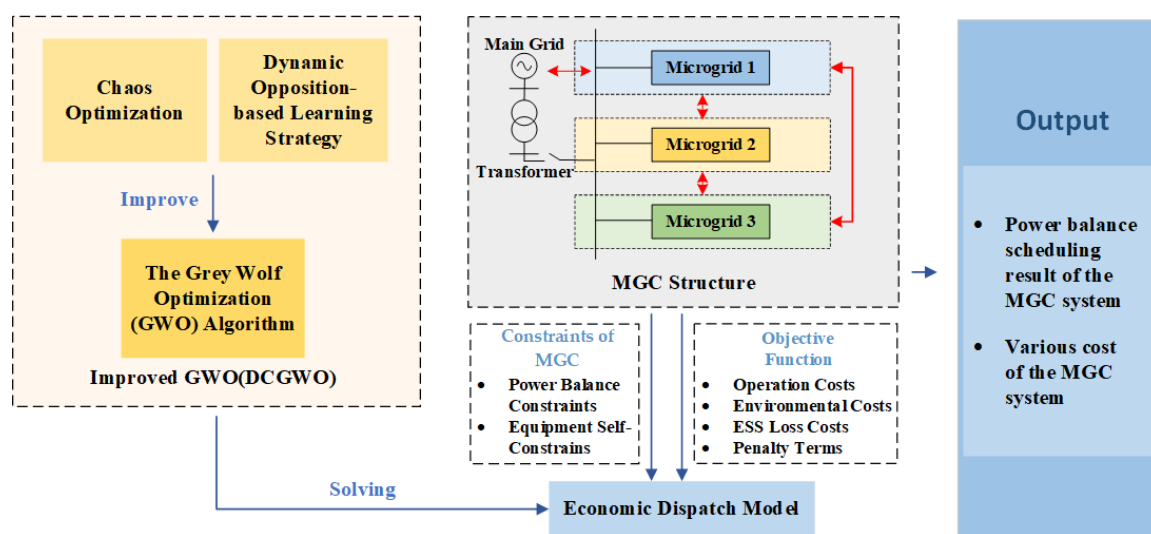


Figure 4. Construction and solution of the MGC economic dispatch model.

4. Simulations and Results

4.1. Parameters of the Numerical Example

This study addresses the optimization scheduling of the MGC system on a daily cycle, with each hour as a scheduling interval, resulting in 24 intervals per day. The meteorological data for the region were sourced from the European Centre for Medium-Range Weather Forecasts (ECMWFs). Using real data from a low-latitude coastal region, including meteorological parameters and historical wind and solar power output data, the output for a typical day is forecasted. The load forecasting data are based on historical data from the local grid.

To calculate the operational revenues of the MGC system, this paper employs time-of-use (TOU) pricing. TOU pricing is a dynamic strategy where electricity rates vary throughout the day, with higher rates during peak demand periods and lower rates during off-peak hours. By incorporating TOU pricing, utilities can encourage consumers to shift

their electricity usage to off-peak times, reducing grid strain during peak periods and optimizing resource utilization. Figure 5 clearly illustrates the electricity purchase and sale prices for each time period throughout the day in the region.

Time-of-Use (TOU) Electricity Pricing for Microgrid Purchase and Sale

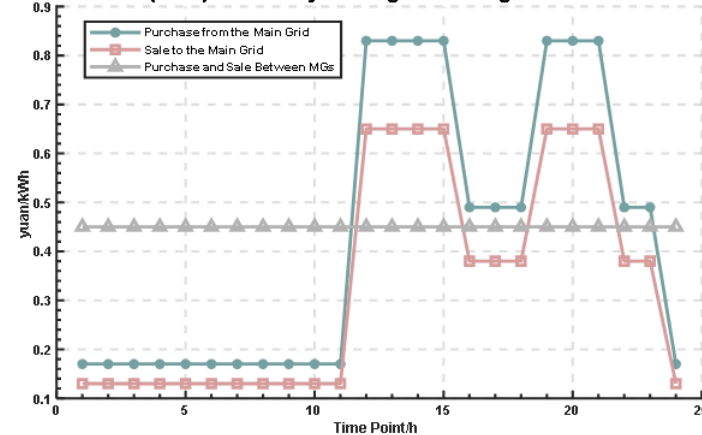


Figure 5. Time-of-use (TOU) electricity pricing for microgrid purchase and sale.

4.2. A Quantitative Analysis of the Effects of Different Chaotic Mappings

In Section 3.2.1, a qualitative analysis was conducted on the compatibility of various chaotic maps with the research context described in this paper. It was concluded that Chebyshev mapping and Logistic mapping are relatively more suitable for optimizing the GWO algorithm. To determine the most effective type of chaotic mapping for the economic dispatch optimization of the MGC, this study integrates the four commonly used chaotic maps with GWO. The resulting four improved chaotic GWO (CGWO) algorithms, along with the unmodified traditional GWO, are applied to the economic dispatch optimization of the MGC system. Their optimization effects are compared, and the comparative convergence characteristic curves are illustrated in Figure 6.

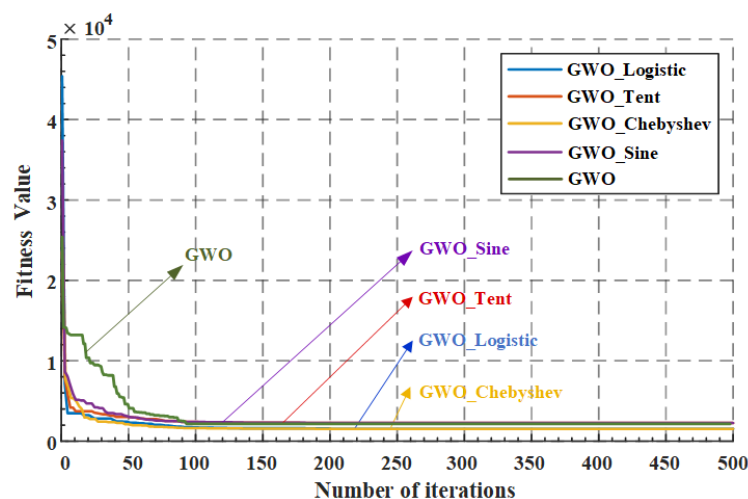


Figure 6. Comparison of convergence characteristic curves for four CGWO variants and traditional GWO.

It can be observed that different chaotic mappings significantly influence the convergence speed of the optimization algorithm in its early iterations. Among the four commonly used chaotic mappings discussed in this paper, CGWO improved with Sine mapping demonstrated the poorest performance, initially exhibiting the slowest convergence speed. Tent mapping also showed suboptimal optimization results, with both the convergence speed and precision falling short. In contrast, the GWO algorithm optimized with Chebyshev mapping produced solutions closest to the actual optimum. However,

Logistic mapping achieved the fastest convergence, with its acceleration effect being particularly notable within the first 20 iterations. To further compare the optimization effects of these two chaotic mappings, a detailed analysis is provided in Table 4.

Table 4. Comparison of the effects of various chaotic mappings.

Type of Chaotic Mapping	Optimal Fitness Value	Runtime/s
Tent	2.0920×10^3	9.1720009
Sine	2.0597×10^3	10.496083
Chebyshev	1.6895×10^3	11.616561
Logistic	1.7150×10^3	7.9097413
Traditional GWO	1.9030×10^3	10.534876

Upon analyzing Table 4, it is evident that while the GWO algorithm optimized with Chebyshev mapping achieves results closer to the actual optimum, the difference compared to the GWO algorithm optimized with Logistic mapping is negligible. Considering that Chebyshev mapping requires significantly more computational time than Logistic mapping, and with the aim to strike a balance between algorithm efficiency and precision, the superior optimization effect of Logistic mapping is asserted.

Building on the comprehensive analysis that includes both qualitative and quantitative evaluations, it is established that Logistic mapping exhibits superiority over other mappings in enhancing the performance of the GWO algorithm. Consequently, this study adopts Logistic mapping to optimize the GWO algorithm.

4.3. Analysis of Simulation Results

4.3.1. An Analysis the Optimization Effect of the Proposed CDGWO

In Section 4.2, experiments were conducted to compare the optimization effects of various chaotic mappings on the GWO algorithm, ultimately selecting Logistic mapping as the most effective overall. However, the convergence characteristic curves of CGWO indicate that while chaotic mapping significantly accelerates convergence in the early stages, its impact diminishes in the later stages. Therefore, this study also incorporates the dynamic opposition-based learning strategy to further enhance the convergence speed and global search capability of the GWO algorithm.

To verify the optimization performance of the improved CDGWO algorithm, CDGWO, traditional GWO, and three other commonly used intelligent optimization algorithms—the Firefly Algorithm (FA), Particle Swarm Optimization (PSO), and Whale Optimization Algorithm (WOA)—were applied to the economic dispatch of the MGC system. The optimization results were then compared and analyzed through charts and graphs. Figure 7 illustrates the convergence characteristic curves of five intelligent optimization algorithms.

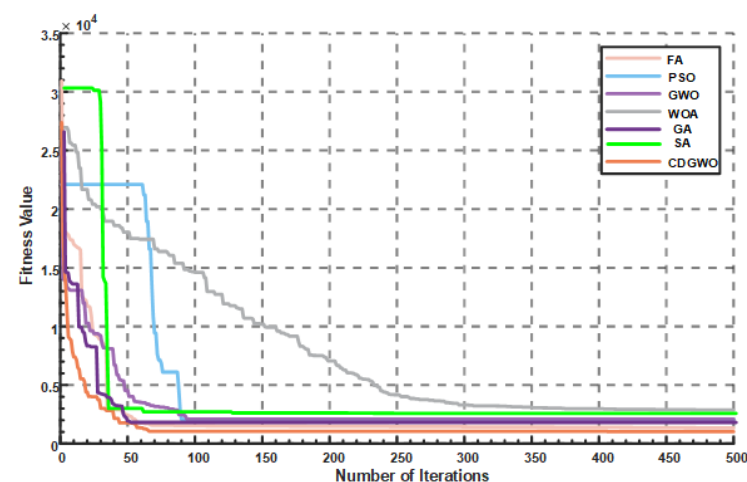


Figure 7. Comparison of improved GWO (CDGWO) with common intelligent optimization algorithms.

For a more comprehensive comparison of algorithm performance, Table 5 provides precise numerical indicators, including the Optimal Fitness Value, Runtime, Number of Iterations at Convergence, and convergence variance.

Table 5. Comparison of improved GWO (CDGWO) with common intelligent optimization algorithms.

Intelligent Optimization Algorithm	Optimal Fitness Value	Runtime/s	Number of Iterations at Convergence	Convergence Variance
FA	1.425×10^3	1100.305307	70	132.656897
PSO	2.147×10^3	27.753490	90	213.926534
WOA	3.045×10^3	26.265349	350	587.452367
GWO	1.903×10^3	10.534876	93	196.567398
GA	1.576×10^3	103.635457	56	206.875623
SA	2.803×10^3	52.786543	130	670.246676
CDGWO	1.044×10^3	6.906439	65	48.678354

Convergence variance reflects the average squared differences among optimization results from multiple runs. A lower convergence variance indicates more consistent results across runs, suggesting algorithm stability. Conversely, a higher convergence variance suggests greater variability, possibly due to algorithm randomness or instability. Analyzing the convergence variance enables an assessment of the optimization algorithm's stability and consistency.

$$S_{Con}^2 = \frac{1}{N} \sum_{i=1}^N (F_i - \bar{F})^2 \quad (21)$$

where N denotes the number of runs, F_i represents optimal fitness value of the i -th run, and $\bar{F}$ denotes the mean of all optimization results.

Figure 7 should be analyzed in conjunction with Table 5. The Whale Optimization Algorithm (WOA) exhibits the slowest convergence speed, needing about 350 iterations to reach the global optimum. This slow convergence is due to its simulation of whale foraging behaviors, which introduces uncertainty in the search direction [52]. Additionally, the WOA has a high convergence variance of 587.45, indicating instability.

In contrast, the Firefly Algorithm (FA) converges more quickly, reaching the global optimum in about 70 iterations, with a lower convergence variance below 150, indicating better stability. However, its computational complexity ($O(n^2)$), due to interactions between fireflies, results in significantly longer computation times compared to other algorithms [53,54].

The Genetic Algorithm (GA) achieves a similar solution accuracy to the FA but is less computationally efficient, with a runtime of 103 s. This is due to the resource-intensive nature of genetic operations, which require significant computational resources.

Simulated Annealing (SA) has a lower solution accuracy and a runtime of 53 s. Its high convergence variance of 670.25 reflects significant instability and a tendency to converge to local optima due to its probabilistic approach.

Both Particle Swarm Optimization (PSO) and traditional Grey Wolf Optimization (GWO) show good stability with a convergence variance of around 200. However, PSO often gets trapped in local optima due to rapid early convergence and limited disturbance mechanisms.

Among the mentioned algorithms, GWO performs reasonably well. However, compared to traditional GWO, CDGWO stands out with improved efficiency and enhanced search capabilities. By the 65th iteration, CDGWO converges to the optimal fitness value, achieving results closest to the actual global optimum among the five algorithms, with the highest precision. Additionally, CDGWO's computation time is 3.6 s shorter than traditional GWO, marking a significant efficiency boost. Moreover, CDGWO demonstrates a notably increased stability, with a convergence variance of only 48.678354, surpassing other algorithms.

In conclusion, the improved GWO algorithm achieves optimization in iteration efficiency, precision, and stability. Among the commonly used intelligent optimization

algorithms discussed in this paper, the CDGWO algorithm demonstrates the best optimization performance. It is worth noting that the inclusion of penalty terms in the objective function means that the fitness value and the actual daily cost of the MGC system, as obtained by various optimization algorithms, are not numerically equivalent. Table A1 presents the actual daily costs obtained by each algorithm. Remarkably, CDGWO achieves the lowest cost, further affirming the superiority of this enhanced algorithm.

4.3.2. Analysis of Economic Dispatch Results Based on CDGWO

In this section, the previously proposed CDGWO algorithm is applied to solve the MGC scheduling model described in Section 4.1. To further analyze the effectiveness of the economic dispatch and verify the stability of the proposed optimization method, comparative experiments were conducted. Initially, forecast data based on actual renewable energy generation and load profile data from a low-latitude coastal region were used as inputs under normal conditions. Additionally, a $\pm 10\%$ random disturbance was introduced to MG1's wind power output, MG2's photovoltaic output, and MG3's load forecast to simulate uncertainties.

In Figures 8–10, (a) represents the optimal scheduling results for each sub-microgrid under normal conditions, and (b) represents the power balance scheduling results for each sub-microgrid after introducing random disturbances. Appendix A includes Tables A2 and A3, which present the power purchase and sale situation of each MG under normal conditions and after introducing random disturbances, respectively. Additionally, Tables A4 and A5 show the various costs of the MGC system under normal conditions and after introducing random disturbances. These figures and tables allow for an analysis from multiple perspectives.

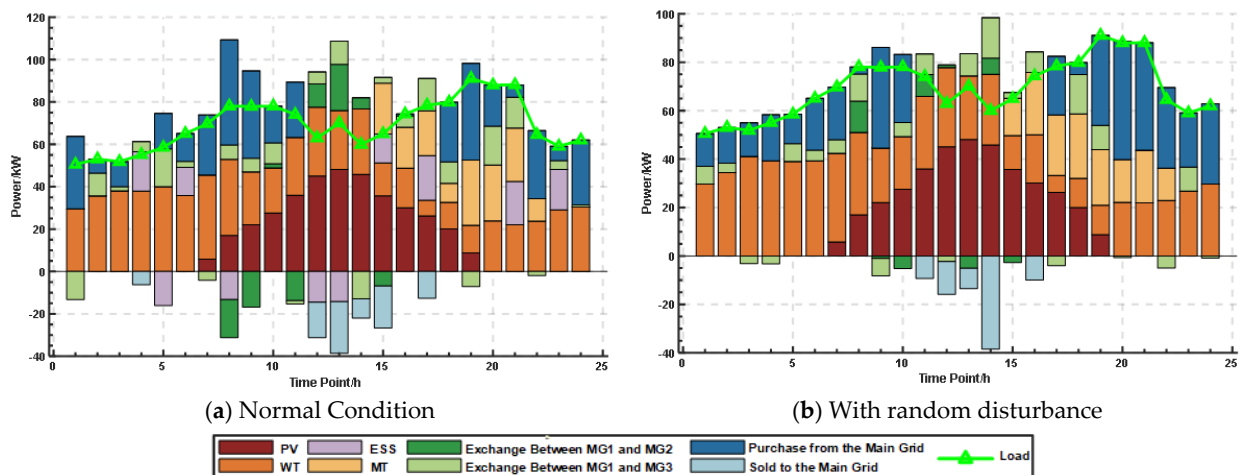


Figure 8. Results of power balance scheduling of MG1.

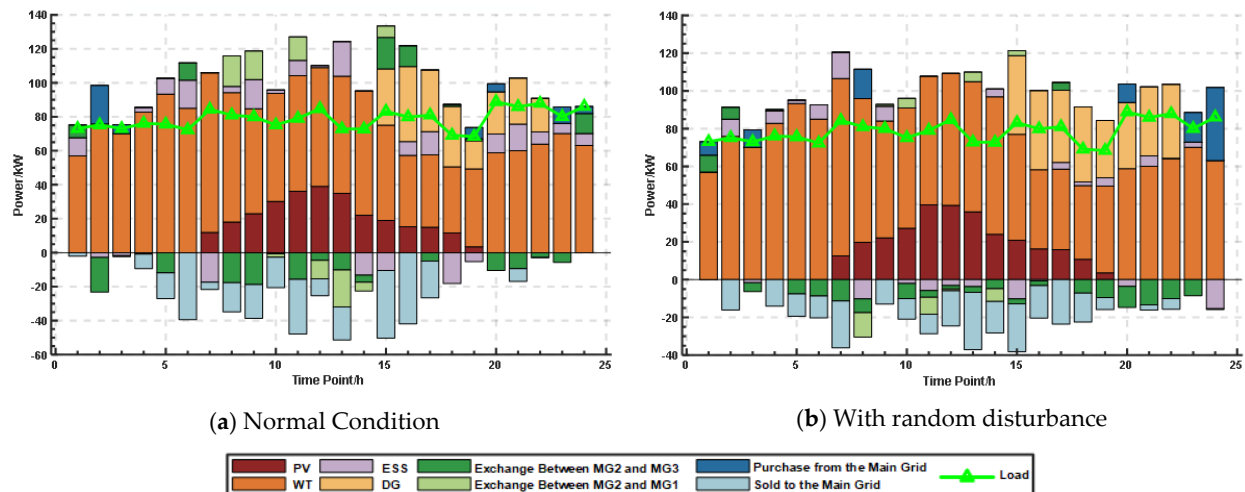


Figure 9. Results of power balance scheduling of MG2.

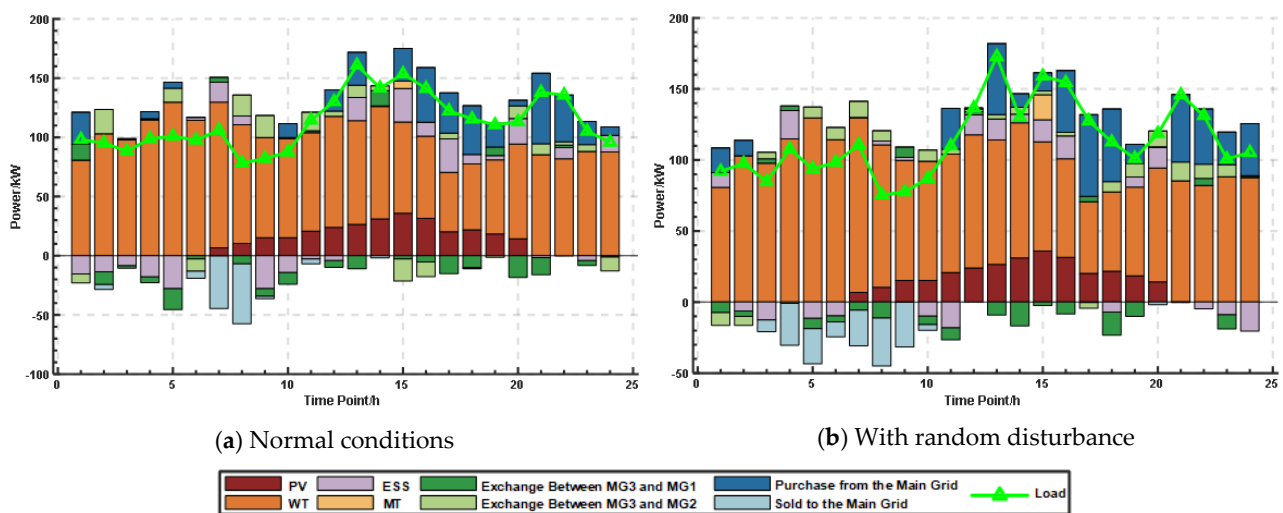


Figure 10. Results of power balance scheduling of MG3.

1. **ESS.** From the standpoint of energy storage, the ESS effectively plays a vital regulatory role, achieving peak shaving and valley filling through flexible charging and discharging. Moreover, the ESS aims to maintain energy balance throughout its operational cycle, which helps extend its lifespan.
2. **Economic Efficiency.** Based on the time-of-use pricing shown in Figure 5, an economic analysis reveals that the three MGs prefer to purchase electricity from the main grid during periods of low prices and sell electricity back to the grid during high-demand periods (at around 1 PM and 8 PM) when prices are higher. This strategy, combined with the ESS discharging and energy exchange between MGs, ensures internal power balance and cost efficiency.
3. **Environmental Protection and Power Quality.** In terms of environmental protection and power quality, all three MGs prioritize utilizing their internal renewable energy sources. When renewable energy is insufficient, traditional thermal power generation is employed for peak shaving and valley filling, ensuring a balanced approach to energy management.
4. **Robustness.** A comparison of the optimization results under normal conditions and with disturbances revealed that the three microgrids maintain perfect power balance, ensuring the power quality of the system, as shown in Figures 8–10. According to Tables A2–A5, the operational costs, pollution control costs, and energy storage system loss costs all increase under disturbances, with the overall costs rising by 7.80%.

The variation is not significant. These results indicate that although the optimization results change slightly after introducing disturbances, the system performance remains stable and ensures excellent economic benefits, demonstrating the reliability and robustness of the method in practical applications.

5. Conclusions

Based on real wind and solar power outputs and load data from a low-latitude coastal region, this paper conducts a comprehensive study on the economic dispatch optimization of microgrid cluster (MGC) systems. This study begins by presenting the topology and equipment configuration of the MGC. Within this framework, it provides a detailed discussion of the self-constraints of each microgrid (MG), the constraints of the energy storage system (ESS), and the power balance constraints within the MGC system. This ensures that the optimization results for the economic dispatch of MGCs are feasible in principle, economically rational, and reliable in system operations. Accordingly, a more optimal multi-objective economic dispatch model is constructed and solved using the improved CDGWO algorithm. Finally, through comparative experiments, the superiority of the proposed CDGWO is comprehensively and specifically analyzed in graphical and tabular form. The robustness of the model is also validated by analyzing the economic dispatch results before and after introducing random disturbances. The main contributions of this study are as follows:

1. ***Incorporation of Penalty Terms.*** In addition to economic indicators, two penalty terms are introduced into the objective function of the multi-objective economic dispatch model: the penalty for power-exchange-exceeding limits between the main grid and the MGC and the penalty for discrepancies in energy levels between the start and end of the ESS operating cycle. This integration minimizes costs while incorporating critical factors such as power quality and equipment lifespan, promoting a more reliable, efficient, and sustainable operation of the MGC.
2. ***Proposed Improved GWO (CDGWO).*** The proposed CDGWO combines chaotic mapping and the dynamic opposition-based learning strategy with traditional GWO. Through experiments, the most suitable type of chaotic mapping for the research context, Logistic mapping, was identified. The improved GWO algorithm achieves significant enhancements in optimization performance.

Overall, the integration of CDGWO with the comprehensive multi-objective optimization model presents a robust solution for the economic dispatch of MGCs, offering both economic benefits and enhanced system reliability. Future research should explore further refinements to this optimization algorithm and its objective function, as well as its application to larger and more complex microgrid systems.

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Appendix A

Table A1. Actual daily cost of MGC solved by various intelligent optimization algorithms.

Intelligent Optimization Algorithm	Daily Cost/CNY
FA	898.42
PSO	1003.21
WOA	1343.77
GWO	937.86
CDGWO	780.46

Table A2. Power purchase and sale situation of each MG under normal conditions.

MG1	Power/kW	MG2	Power/kW	MG3	Power/kW
Total power purchased from the main grid	414.8024	Total power purchased from the main grid	53.5579	Total power purchased from the main grid	396.2862
Total power sold to the main grid	88.9414	Total power sold to the main grid	297.6848	Total power sold to the main grid	113.1721
Net power exchange between MG1 and MG2	15.4243	Net power exchange between MG2 and MG3	75.4065	Net power exchange between MG3 and MG1	108.5003
Net power exchange between MG3 and MG1	108.5003	Net power exchange between MG1 and MG2	15.4243	Net power exchange between MG2 and MG3	75.4065

Table A3. Power purchase and sale situation of each MG after introducing random disturbances.

MG1	Power/kW	MG2	Power/kW	MG3	Power/kW
Total power purchased from the main grid	437.9258	Total power purchased from the main grid	96.2172	Total power purchased from the main grid	442.5331
Total power sold to the main grid	79.5980	Total power sold to the main grid	274.2019	Total power sold to the main grid	169.3209
Net power exchange between MG1 and MG2	15.8488	Net power exchange between MG2 and MG3	105.5076	Net power exchange between MG3 and MG1	100.7911
Net power exchange between MG3 and MG1	100.7911	Net power exchange between MG1 and MG2	15.8488	Net power exchange between MG2 and MG3	105.5076

Table A4. Various costs of the MGC system under normal conditions.

Operational Cost	CNY	Pollution Control Cost	CNY	ESS Lose Cost	CNY
MG1	146.3235	MG1	44.1442	MG1	0.0022
MG2	94.7552	MG2	90.4444	MG2	0.0030
MG3	403.1297	MG3	1.6524	MG3	0.0033
Total	644.2084	Total	136.2411	Total	0.0085
Total cost			CNY 780.46		

Table A5. Various costs of the MGC system after introducing random disturbances.

Operational Cost	CNY	Pollution Control Cost	Yuan	ESS Lose Cost	CNY
MG1	182.8861	MG1	44.6929	MG1	0.0002
MG2	105.6780	MG2	115.3801	MG2	0.0017
MG3	388.0392	MG3	4.6675	MG3	0.0031
Total	676.6033	Total	164.7405	Total	0.0049
Total cost			CNY 841.35		

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